DL-Ops Lab Assignment 1

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1 Introduction

This report presents experiments performed on MNIST and FashionMNIST datasets using deep learning and machine learning models. The objective is to study the effect of batch size, optimizer, learning rate, and computing platform (CPU/GPU) on model performance.

2 Q1(a): Deep Learning Experiments

2.1 MNIST Results

Table 1: MNIST Test Classification Accuracy (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	98.97	98.86
16	SGD	0.0001	96.37	94.77
16	Adam	0.001	98.87	98.71
16	Adam	0.0001	**99.17**	98.36
32	SGD	0.001	98.38	98.36
32	SGD	0.0001	NA	NA
32	Adam	0.001	NA	NA
32	Adam	0.0001	99.08	98.81

2.2 FashionMNIST Results

Table 2: FashionMNIST Test Classification Accuracy (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	90.42	89.64
16	SGD	0.0001	83.36	76.22 (3 ep) / 86.39 (8 ep)
16	Adam	0.001	90.59	89.22
16	Adam	0.0001	**91.62**	90.20
32	SGD	0.001	89.27	87.89
32	SGD	0.0001	NA	NA
32	Adam	0.001	NA	NA
32	Adam	0.0001	91.04	89.41

2.3 Observations

- Adam optimizer provided better accuracy compared to SGD in most cases.
- Lower learning rate (0.0001) resulted in better convergence for both datasets.

- ResNet-18 achieved performance comparable to or better than ResNet-50.
- FashionMNIST was more challenging compared to MNIST, leading to lower accuracy.
- In most of the experiments 3 epochs were giving good accuracy for one of the configuration I used 8 epoch to get accuracy of more that 80
- The training vs validation graph is shown (figure 1 and figure 2) for the best configuration i.e with batch size=16, optimizer=Adam,learning rate =0.0001 and model= resnet18

3 Q1(b): SVM Classification

SVM classifiers were trained on MNIST and FashionMNIST using polynomial and RBF kernels. Training time was measured in milliseconds.

Table 3: SVM Results

Dataset	Kernel	Accuracy (%)	Training Time (ms)
MNIST	Polynomial	91.70	4578
MNIST	RBF	93.15	4172
FashionMNIST	Polynomial	81.55	3928
FashionMNIST	RBF	85.15	3791

3.1 Observations

- RBF kernel achieved higher accuracy than polynomial kernel.
- SVM training time was significantly higher compared to neural networks.
- SVM performed better on MNIST than FashionMNIST.

4 Q2: CPU vs GPU Performance

Table 4: CPU vs GPU Performance Comparison

Compute	Batch Size	Optimizer	Learning Rate	Test Classification Accuracy (%)			Train Time (ms)			FLOPs		
				ResNet-18	ResNet-32 [Optional]	ResNet-50	ResNet-18	ResNet-32 [Optional]	ResNet-50	ResNet-18	ResNet-32 [Optional]	ResNet-50
CPU	16	SGD	0.001	85.95	NA	80.62	54075	NA	130139	1.824	NA	4.132
CPU	16	Adam	0.001	85.85	NA	80.95	56384	NA	131496	1.824	NA	4.132
GPU	16	SGD	0.001	90.42	NA	89.64	40988	NA	76061	1.824	NA	4.132
GPU	16	Adam	0.001	90.59	NA	89.22	44224	NA	81390	1.824	NA	4.132

4.1 Observations

- GPU training significantly reduced training time compared to CPU.
- FLOPs remained constant across CPU and GPU, indicating FLOPs depend on architecture.
- ResNet-50 required more computation time compared to ResNet-18.
- Dataset size and image resolution were reduced for CPU experiments to ensure feasibility.

5 Training and Validation Curves

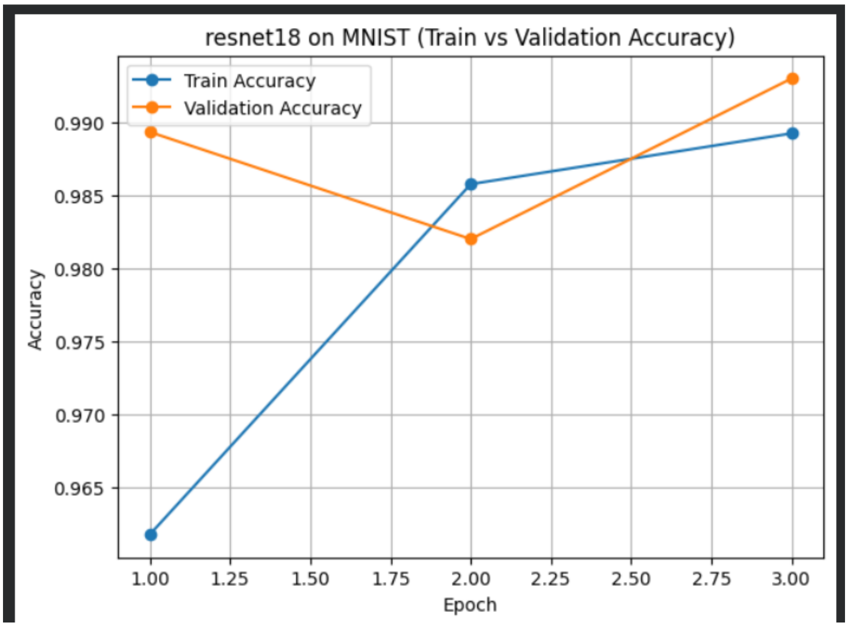

Figure 1: Training vs Validation Accuracy

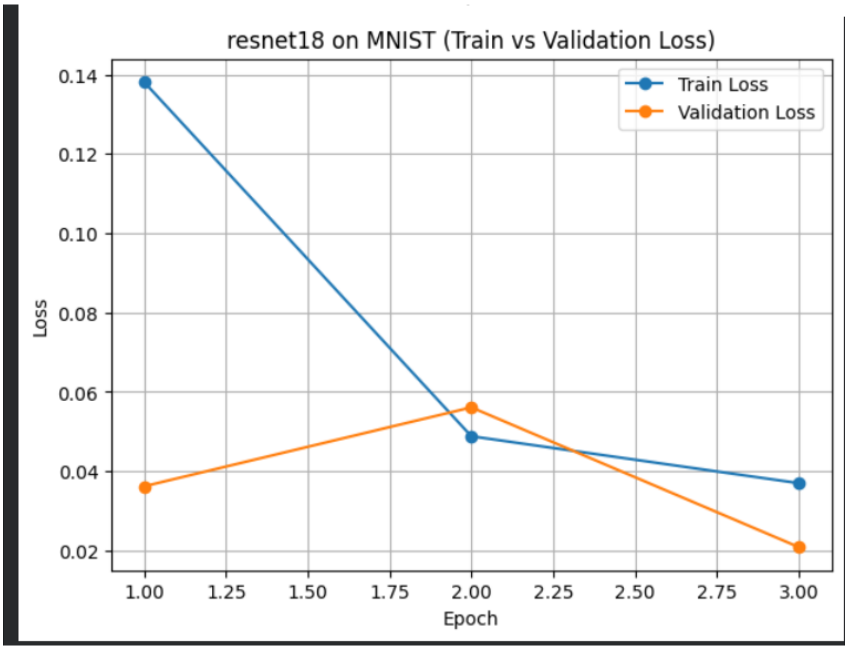

Figure 2: Training vs Validation Loss

6 Conclusion

The experiments show that optimizer choice, learning rate, and compute platform significantly affect model performance. ResNet-18 provides a good balance between accuracy and computational cost. GPU acceleration is essential for training deeper models efficiently.

Links

- Colab Notebook: https://colab.research.google.com/drive/14PQwbYVtsiNgUgT1_LfDSVID2RHZJTuJ?usp=sharing
- GitHub Repository: https://github.com/KartikIyer27/MlOps-kartik-m25CSA025/tree/Assignment-1